

Branch Comparison Report

MEP-AUTOMATION-TOOLS vs develop | CED_Extensions repository | Prepared 2026-07-17 | Reed Pinterich / CoolSys Energy Design

Purpose: Inventory of everything the **MEP-AUTOMATION-TOOLS** branch contains that **develop** does not, to support a decision on re-branching off develop into a clean feature branch and retiring this branch.

166 commits ahead of develop (Feb 2 – Jul 16, 2026)	863 changed paths total	790 brand-new files (additive)	51 modified shared files (merge-risk zone)	107 commits develop has that this branch lacks
---	---	--	--	--

Branch-health note: Git reports *multiple merge bases* between the two branches (comparison used `c78e99a`, 2026-04-21, "Targeted Bug Fixes for Name Piping Systems"). The history contains reverted merges and self-merges — consistent with the decision to rebuild on a clean branch.

1. MEPRFP Automation 2.0 Panel — entire panel, absent from develop (197 files)

The flagship. Develop has **none** of it.

- **Circuiting pulldown** — SuperCircuit V5 (modeless preview, client-pluggable rules), Audit Circuits
- **Place from CAD or Linked Model** — parent-matching placement of full profiles from linked Revit/CAD/CSV
- **Place Single Profile** (2.0 variant), **Place Element Annotations**, **Check Synced Relationships**
- **Spaces pulldown** — space/bucket-based profile system with door-aware placement rules
- **Modify Profiles** and **Import-Export Profiles** pulldowns
- **Misc Ops pulldown** — including the nine-category QAQC drift audit
- **lib/** — ~100-module engine: profile/space models, directives (BYPARENT/BYSIBLING), Element_Linker lineage, Extensible Storage v4 codec, YAML round-trip I/O, schema migrations, client circuiting adapters (HEB / PF), plus the pure-logic test suite and runner

2. MEP Automation Panel (1.0) — major expansion (299 files added)

Develop holds only a thin, pre-February version of this panel. Added since:

- **Audit Tools pulldown:** QAQC, Audit CAD Model, Follow Parent, Hide Existing Profiles, Place Identity Mark, Post Audit Compare, Where da stuff
- **Electrical pulldown:** SuperCircuit V3 / V4 / V5 (color-coded circuit preview), AI SuperCircuitV5 + headless runner, Arrange Lights in Space, CircuitBySpace, Color fixture by dev group, Color fixture by Host Name, Refrigeration Tag, Audit Circuits

- **Misc Operations pulldown:** Optimize (nudge engine), Update Profiles, Update Vector, Place Annotations, Place Spaces CSV, Normalize Truth Groups, Audit Linked Model 1.0 & 2.0, Audit CAD Model 2.0 ("batman"), Convert Schema, SchemaViewer, Write Element Linker, Copy Import Layers
- **Space Operations pulldown:** Classify Spaces, Manage Space Profiles, Place Space Elements, Place Space Annotations, Import / Export Space Configs
- **Modify Profiles stack:** Modify Profiles, Place Items, and Import-Export pulldowns (Add / Merge / Bulk Merge / Unmerge / Delete / Edit-Create / Manage Profiles, Place CAD / Linked Elements, Remove Profiles)
- **Standalone:** Place Single Profile dockable pane, Create Independent Profile, Check Synced Relationships

3. QualityChecks Panel — entire panel, absent from develop (44 files)

- Proximity Check (lights vs coils, sync-triggered), General Proximity Check, Ref Sched Change watcher, Flag CC Circuits, AI_QC pulldown (AI-developed QC check suite)

4. CED ElecTools Extension

- **New tools:** Circuit Grouper (full tool: cg_core / cg_collect / cg_apply / cg_window + unit tests; group-by-space, group-by-parameter, auto panel selection) and CC Load Injection (with optional load balancing) — the most recent July work
- **Heavily modified:** Circuit Manager / CircuitBrowserPanel (+ new Neutral/IG action window), Let there be Light, Add Spares and Spaces, BatchSwapCircuits, Edit Circuit Properties, Alerts Manager, startup.py
- **Deleted here:** Circuit Element Finder button and its CEDLib service/domain modules

5. CED MechTools Extension

- **New:** Category Converter button, RefOps shared-parameter definitions
- **Modified:** System Tagger (Identity Mark lookup change), Name Piping Systems, Print Pipe Data, SetPipeSystem

6. H-E-B Tools Extension & FamilyOperations

- **H-E-B Tools (new):** Project Mates button (bundled app.exe + msedgedriver), Word Report / revision report generator (script + exe), extension manifest
- **FamilyOperations panel:** Product Injector button (rest of panel already on develop)

7. MCP Automation (non-pyRevit)

- **Planet Fitness_MCP_AUTOMATION** (79 files): Electrical Power_auto training dump, SYNTHESIS.md, PF circuiting algorithm, power-plan playbook, collect/replicate skills
- **HEB_MCP_AUTOMATION** (66 files): bakery_auto, pharmacy_auto, collect/replicate skill scripts, README

8. Data Assets & Root Utilities

- **Profile stores:** HEB_profiles V4_MODIFIED_30/36/37/40/41, HEB_profiles V5 / 5.1–5.4 / 5.8 series + dedupe/backup copies, PF_profiles_V4_23 (+ .bak), PF_CorporateONLY power profiles, WFM_Profiles_1, WFMELECPROFILES, WFMWETProifles2, WFMWETrefrProifles1, Test_HEB_Space_profile_4
- **Shared parameters:** AE CoolSys Energy Design (CED) Shared Parameters.txt
- **YAML transform scripts:** _convert_v4_to_v5, _dedupe_v5_profiles, _make_v5_4, _patch_v5_is_group, fix_parent_directives_V5_2, build_HEB_profiles_V5_1
- **Docs / decks:** MEPRFP_Automation_2_Overview (py + pptx), PF_Revit_Automation deck + backup, append_profile_slides.py, CED_Software_Enhancements_Summary.xlsx, CLAUDE.md, .claude/CLAUDE.md

9. Modified Shared Code — the merge-risk zone (51 files)

These are edits to files develop *also* has (and develop has moved 107 commits since the fork), so they are the only real conflict surface:

Area	Files
CEDLib.lib / LogicClasses	placement_engine.py, profile_schema.py, profile_repository.py, csv_helpers.py, yaml_path_cache.py, __init__.py
CEDLib.lib / ExtensibleStorage	extensiblestorage.py, yaml_store.py, workflow notes
CEDLib.lib / UIClasses	PlaceElementsUI, ProfileEditorWindow (py + xaml), ProfileSelectionWindow, SearchBox.xaml, resource_loader, revit_theme_bridge, ui_bases
CEDLib.lib / CEDElectrical	set_include_and_recalculate_operation.py, circuit_properties_editor.py
CEDLib.lib / pyrevitmep	manageproject.py, manageshared.py
Startup / config	AE pyTools startup.py, CED ElecTools startup.py, .gitignore, .claude/settings.local.json, tab/panel bundle.yaml files
Other extensions	ElecTools circuit tools (6 scripts), MechTools Ref Ops (4 scripts), WM Tools Load Electrical Content

10. Deletions on this branch (21 files)

- Circuit Element Finder pushbutton + its four CEDLib modules
- docs/release_2_0_description.md and release_2_0_devnotes.md
- CEDLib AI_Agents planning docs (Planner.md, LOCK.md, open_questions.md, 3 documentation files)
- Copy Import layers pushbutton (old location — re-added under Misc Operations), RefrigerationIndependentProfiles.yaml, two icon sets

11. Uncommitted Working-Tree Changes (as of 2026-07-17)

Not yet in any commit — must be committed or stashed before any branch surgery, or they will be lost:

- pyRevit 6.5.3 `setLevel` compat shims: SuperCircuitV4, SuperCircuitV5 (script + bundle.yaml), AI SuperCircuitV5 (script + headless_run)
- Place Single Profile crash fix: refresh now routed through ExternalEvent (PlaceSingleProfilePanel.py)
- startup.py cleanup: removed undefined `_register_circuit_browser_panel()` call (NameError on every boot since March) and stale disable comments
- SuperCircuit V5 (MEPRFP 2.0) color-coded grouping port: circuit_window.py + SuperCircuitWindow.xaml
- Untracked: pyRevit_653_Impact_Audit_2026-07-16.pdf

12. Assessment for the Re-branch Plan

~92% of the delta is purely additive (790 of 863 paths are new files develop has never seen) — these carry to a fresh branch conflict-free. The 51 modified shared files are the only genuine merge-conflict surface against develop's 107 newer commits. Recommended order: (1) commit the uncommitted fixes on this branch first so all work is recorded; (2) branch off develop; (3) bring the additive file set over wholesale; (4) hand-merge the 51 shared-file edits; (5) retire MEP-AUTOMATION-TOOLS only after the new branch builds and the flagship tools verify in Revit.

Generated from: `git log develop..MEP-AUTOMATION-TOOLS / git diff --name-status develop...MEP-AUTOMATION-TOOLS` (merge base c78e99a, 2026-04-21) / `git status`. Local develop is in sync with origin/develop (0/0 ahead-behind).